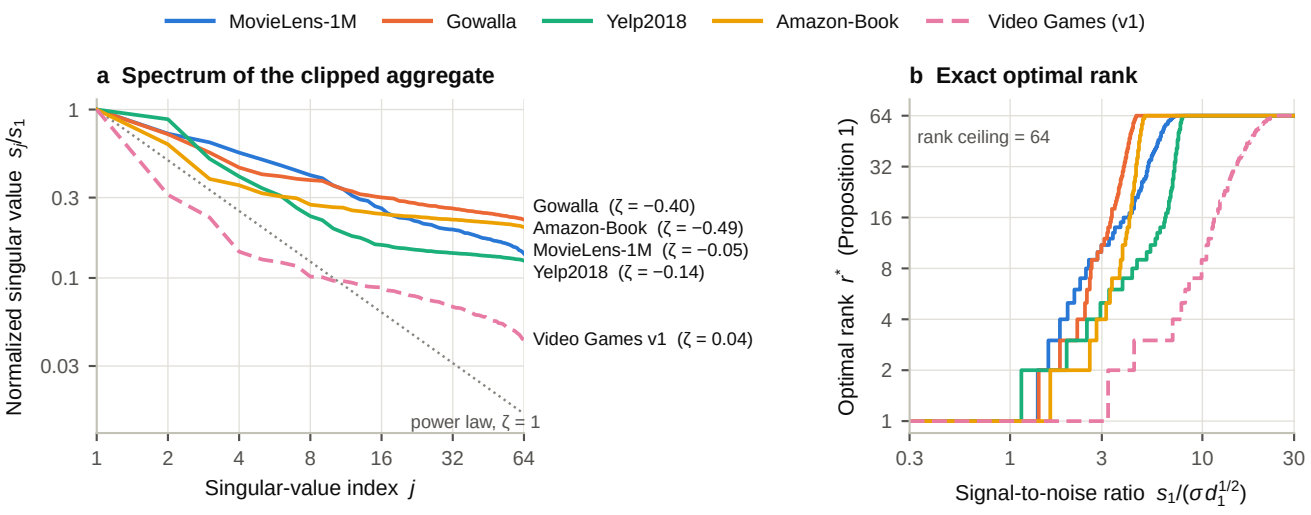
MovieLens-1M
Gowalla
Yelp2018
Amazon-Book
Video Games (v1)
a Spectrum of the clipped aggregate
Normalized singular value s_j/s_1
1
0.3
0.1
0.03
Gowalla (ζ = −0.40)
Amazon-Book (ζ = −0.49)
MovieLens-1M (ζ = −0.05)
Yelp2018 (ζ = −0.14)
Video Games v1 (ζ = 0.04)
power law, ζ = 1
1
2
4
8
16
32
64
Singular-value index j
b Exact optimal rank
Optimal rank r^* (Proposition 1)
64
32
16
8
4
2
1
rank ceiling = 64
0.3
1
3
10
30
Signal-to-noise ratio $s_1/(\sigma d_1^{1/2})$